

Interview Q&A: machine learning

1. What is the difference between supervised and unsupervised learning?

Supervised learning involves training a model on labeled data to make predictions, while unsupervised learning involves discovering patterns or relationships in unlabeled data.

2. What is the purpose of normalization in machine learning?

Normalization is used to scale input features to a common range, typically between 0 and 1, to prevent features with large ranges from dominating the model.

3. What are some common types of regression algorithms?

Common types of regression algorithms include Linear Regression, Ridge Regression, Lasso Regression, Elastic Net Regression, and Polynomial Regression.

4. What is the bias-variance tradeoff in machine learning?

The bias-variance tradeoff refers to the balance between a model's ability to fit the training data well (low bias) and its ability to generalize to new, unseen data (low variance). A model with high bias will not fit the training data well, while a model with high variance will overfit the training data.

5. What is the concept of overfitting in machine learning?

Overfitting occurs when a model is too complex and learns the noise in the training data, resulting in poor performance on new, unseen data.

6. What is the difference between a neural network and a deep learning model?

A neural network is a type of machine learning model that can be shallow or deep, whereas deep learning specifically refers to a subset of machine learning models that use multiple layers to learn complex representations.

7. What is the role of hyperparameter tuning in machine learning?

Hyperparameter tuning involves adjusting parameters such as learning rate, regularization strength, or number of hidden layers to optimize model performance on a given task.

8. What is the concept of feature engineering in machine learning?

Feature engineering is the process of selecting, transforming, and creating new features from existing ones to improve the performance of a machine learning model.

9. What is the purpose of using cross-validation in machine learning?

Cross-validation is used to evaluate a model's performance on unseen data by splitting the available data into training and testing sets, and then repeating this process multiple times.

10. What is the difference between a model's accuracy and precision?

Accuracy measures the proportion of correct predictions, while precision measures the proportion of true positives among all positive predictions.

11. What is the concept of dimensionality reduction in machine learning?

Dimensionality reduction is a technique used to reduce the number of features or dimensions in a dataset while retaining most of the information, making it easier to analyze and visualize.

12. What is the difference between a model's loss function and its objective function?

A model's loss function measures the difference between predicted and actual outputs, while its objective function represents the goal of the model to minimize or maximize.

13. What is the difference between a model's training set and its validation set?

The training set is used to train the model, while the validation set is used to evaluate the model's performance during training.

14. What is the difference between a model's training set and its testing set?

The training set is used to train the model, while the testing set is used to evaluate the model's performance on unseen data.

15. What is the purpose of using stochastic gradient descent as an optimization algorithm in machine learning?

Stochastic gradient descent is used to minimize the loss function by iteratively adjusting the model's parameters, allowing for faster convergence and better handling of large datasets.

16. What is the concept of a decision boundary in machine learning?

A decision boundary is an imaginary line or surface that separates different classes or categories in a classification problem.